VIVEKANAND EDUCATION SOCIETY'S INSTITUTE OF TECHNOLOGY

(An Autonomous Institute Affiliated to University of Mumbai
Department of Computer Engineering)

Department of Computer Engineering



Project Report on

Intelligent Tender Management and Contract Lifecycle Platform

In partial fulfillment of the Third Year (Semester–VI), Bachelor of Technology (B.Tech.) Degree in Computer Engineering at the University of Mumbai Academic Year 2025-26

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(2025-26)

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CERTIFICATE of Approval

This is to certify that _____of Third Year Computer Engineering studying under the University of Mumbai has satisfactorily presented the project on "***Intelligent Tender Management and Contract Lifecycle Platform***" as a part of the coursework of CAPSTONE PROJECT - I for Semester-VI under the guidance of ***Dr. Dashrath Mane*** in the year 2025-26.

_____
Date

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Internal Examiner External Examiner

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Project Mentor Head of the Department Principal
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We are thankful to our college Vivekanand Education Society's Institute of Technology for considering our project and extending help at all stages needed during our work of collecting information regarding the project.

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We are deeply indebted to Head of the Computer Department **Dr.(Mrs.) Nupur Giri** and our Principal **Dr. (Mrs.) J.M. Nair ,** for giving us this valuable opportunity to do this project.

We express our hearty thanks to them for their assistance without which it would have been difficult in finishing this project synopsis and project review successfully.

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We wish to express our profound thanks to all those who helped us in gathering information about the project. Our families too have provided moral support and encouragement several times.

Computer Engineering Department

COURSE OUTCOMES FOR CAPSTONE PROJECT

Learners will be to:-

Course Outcomes	Description of the Course Outcomes
CO1	Identify and define a real-world problem through systematic requirement analysis and investigation.
CO2	Conduct a thorough literature review and benchmarking to propose innovative solutions.
CO3	Develop a structured project plan with well-defined milestones, timelines, and risk mitigation strategies.
CO4	Design and implement a working prototype using appropriate engineering tools and methods.
CO5	Analyze experimental results, validate the solution, and optimize performance based on data.
CO6	Effectively document and present the project outcomes, demonstrating professional ethics, teamwork, and project management skills.

ABSTRACT

The procurement and contract management process involves complex bid evaluation, compliance verification, and post-award execution monitoring. Traditional manual tendering systems are prone to delays, human error, limited transparency, and vulnerability to fraudulent practices. This project proposes an I**ntelligent Tender and Contract Management System** that integrates AI-assisted bid evaluation, anomaly detection, post-award monitoring, and historical bidder profiling.

The system assigns bid weights based on officer-defined priorities and detects anomalies such as shell companies, duplicate submissions, abnormally low bids, and suspicious documents. After contract award, the system supports contract-type-specific monitoring. For supply contracts, officers or committee members confirm delivery milestones using checklist-based verification. For work contracts, committees upload incremental progress reports describing milestone completion and observations.

An AI-powered chatbot analyzes tender specifications, contractual clauses, uploaded reports, and timelines to detect delays and recommend applicable penalties. Final decisions remain under human authority, while AI provides data-driven suggestions, risk scores, and a complete digital audit trail. The system aims to enhance transparency, efficiency, accountability, and trust in procurement processes.

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Chapter 1: Introduction

This chapter introduces the concept of the proposed system, highlighting the significance of procurement and tender management in modern organizations. It discusses the motivation behind developing an intelligent system, identifies the drawbacks of existing approaches, defines the problem, explains the relevance of the project, and outlines the methodology used for its development. The chapter establishes the foundation for understanding how Artificial Intelligence can be leveraged to improve transparency, efficiency, and decision-making in tender management systems.

1.1 Introduction to the Project

Procurement and tender management are critical administrative and financial processes in both public and private sector organizations. These processes involve issuing tenders, receiving bids, evaluating proposals, awarding contracts, and monitoring the execution of awarded projects. Due to the involvement of multiple stakeholders, large volumes of data, and complex evaluation criteria, these processes are inherently challenging and require a high degree of accuracy, transparency, and accountability.

Traditional tender management systems primarily rely on manual or semi-automated approaches, which are often inefficient and prone to inconsistencies. With the rapid advancement of digital technologies, there is a growing need to transform these systems into intelligent, automated platforms capable of handling complex decision-making tasks.

The proposed Intelligent Tender Management and Contract Lifecycle Platform is designed to address these challenges by integrating Artificial Intelligence (AI) techniques into the procurement workflow. The system automates bid evaluation using weighted scoring mechanisms and applies machine learning-based anomaly detection techniques to identify suspicious patterns such as abnormally low bids, duplicate submissions, and fraudulent entities.

Furthermore, the system extends beyond bid evaluation by incorporating contract lifecycle management

features, including post-award monitoring, delay detection, and penalty recommendation.

1.2 Motivation for the Project

The motivation for developing this system stems from the increasing complexity and inefficiency of traditional procurement processes. In large-scale organizations, tender evaluation involves analyzing multiple bids based on various parameters such as cost, technical specifications, experience, and delivery timelines. Performing such evaluations manually is not only time-consuming but also susceptible to human bias and error.

Another significant concern is the rise in fraudulent activities within procurement systems. Practices such as submission of fake documents, creation of shell companies, and manipulation of bid values undermine the integrity of the process. Existing systems lack intelligent mechanisms to detect such anomalies effectively.

Additionally, post-award contract monitoring remains a weak area in most systems. Once a contract is awarded, tracking progress, ensuring compliance, and enforcing penalties for delays are often handled manually, leading to inefficiencies and lack of accountability.

The project is therefore motivated by the need to:

- Introduce automation in decision-making processes
- Utilize AI for intelligent analysis and anomaly detection
- Improve transparency and auditability
- Reduce manual effort and operational delays
- Enable data-driven procurement strategies

1.3 Drawback of the Existing System

Existing tender management systems exhibit several limitations that hinder their effectiveness:

a. Manual and Time-Consuming Processes:
   Bid evaluation requires analyzing multiple parameters across numerous submissions, which significantly increases processing time and reduces efficiency.
b. Subjective Decision-Making:
   Human involvement in evaluation introduces bias and inconsistency, affecting fairness and reliability.
c. Lack of Intelligent Fraud Detection:
   Current systems are not equipped with advanced analytical tools to detect anomalies such as duplicate bids, abnormal pricing patterns, or forged documents.
d. Ineffective Post-Award Monitoring:
   Monitoring of contract execution is often unstructured and lacks real-time tracking, resulting in delays and compliance issues.
e. Absence of Historical Data Utilization:
   Previous performance records of bidders are not systematically maintained or utilized, leading to repeated selection of unreliable vendors.
f. Limited Transparency and Auditability:
   The absence of comprehensive digital audit trails makes it difficult to track decisions and ensure accountability.

1.4 Problem Definition

The procurement and tender management process is characterized by complexity, high data volume, and the need for accurate decision-making. However, traditional systems fail to provide efficient, transparent, and intelligent solutions to manage these processes effectively.

There is a need for a system that can automate bid evaluation, detect fraudulent activities, monitor contract execution, and provide decision support based on historical data and real-time analysis.

Problem Statement: To design and develop an intelligent, AI-enabled tender management and contract lifecycle system that automates bid evaluation, identifies anomalies, facilitates post-award monitoring,and enhances transparency, efficiency, and accountability in procurement processes while retaining human control over final decisions.

1.5 Relevance of the Project

The proposed project is highly relevant in the context of digital transformation and the increasing adoption of Artificial Intelligence in organizational processes.

In government sectors, transparent and efficient procurement systems are essential to prevent corruption and ensure optimal utilization of public funds. Similarly, in corporate environments, efficient procurement directly impacts operational costs and project timelines.

The system contributes to:

- Digital Governance: Promotes automation and transparency in public procurement
- Fraud Prevention: Enhances detection of suspicious activities
- Operational Efficiency: Reduces time and effort in evaluation and monitoring
- Data-Driven Decision Making: Utilizes historical and real-time data for improved accuracy
- Scalability and Adaptability: Can be extended to various domains requiring contract management

The integration of AI in procurement systems aligns with global trends in smart governance and intelligent enterprise solutions.

1.6 Methodology Used

The development of the proposed system follows a systematic and structured methodology to ensure efficiency and reliability.

1. Requirement Analysis:

Identification of system requirements based on procurement workflows, stakeholder roles, and functional needs.

2. System Design:

Designing system architecture and modules using diagrams such as block diagrams, use case diagrams, data flow diagrams, and entity-relationship models.

3. Technology Stack Selection:

Selection of appropriate technologies including React.js for frontend, Node.js and Express.js for backend, MongoDB for database management, and Python for AI modules.

4. Implementation of Core Modules:

Development of key system components such as bid evaluation engine, anomaly detection module, monitoring system, and chatbot interface.

5. AI Model Development:

Application of machine learning algorithms for:

- Risk scoring
- Pattern recognition
- Anomaly detection
- Delay prediction and penalty recommendation

6. Integration and Testing:

Combining all modules and performing rigorous testing to ensure system correctness, performance, and reliability.

7. Deployment:

Deployment on cloud platforms to enable scalability, accessibility, and real-time data processing.

Chapter 2: Literature Survey

This chapter presents a comprehensive review of existing research, articles, and technological developments related to intelligent procurement systems and tender management. It includes analysis of relevant research papers, references from books and articles, insights from domain understanding, and a study of existing patents. The objective is to identify current advancements, limitations, and research gaps that justify the need for the proposed system.

2.1 Research Papers – Abstract and Inference

Paper 1: Artificial Intelligence in E-Procurement: A Literature Review (2023)

Abstract:
This paper provides a comprehensive review of the application of Artificial Intelligence in e-procurement systems. It highlights how AI techniques such as machine learning, natural language processing, and predictive analytics are used to automate procurement processes, improve supplier selection, and enhance decision-making. The study emphasizes the role of AI in reducing manual effort, increasing efficiency, and improving transparency in procurement operations.

Inference:
The paper establishes that AI can significantly enhance procurement efficiency and transparency. However, it mainly focuses on general automation and lacks detailed implementation of fraud detection and post-award monitoring. This creates a research gap addressed by the proposed system through anomaly detection and lifecycle management.

Paper 2: The Role of Artificial Intelligence in the Procurement Process: State of the Art and Research Agenda (2023)

Abstract:
This paper discusses the current state of AI adoption in procurement and identifies future research directions. It explains how AI can support decision-making, supplier evaluation, and risk assessment. The study also highlights challenges such as data quality, lack of trust in AI systems, and integration issues with existing workflows.

Inference:
The paper highlights the importance of AI in decision support systems but identifies challenges in real-world implementation. The proposed system addresses these issues by maintaining human oversight, providing

explainable outputs (risk scores), and integrating AI with existing workflows through dashboards and chatbot interfaces.

Paper 3: Deep Learning for Detecting Bid Rigging (2021)

Abstract:
This research focuses on the use of deep learning techniques, specifically convolutional neural networks (CNNs), to detect bid rigging and cartel behavior in procurement systems. The model analyzes bidding patterns and identifies suspicious similarities among bidders. The results demonstrate that AI can effectively detect fraudulent activities that are difficult to identify manually.

Inference:
This paper provides strong evidence for using AI in fraud detection. However, it focuses only on bid rigging detection and does not cover other aspects such as evaluation scoring or contract monitoring. The proposed system extends this concept by incorporating broader anomaly detection along with evaluation and monitoring functionalities.

Paper 4: Artificial Intelligence-Based Process Automation in E-Procurement: A Systematic Literature Review (2022)

Abstract:
This paper reviews various AI-based automation techniques used in e-procurement systems. It discusses methods for supplier selection, process optimization, and performance evaluation. The study highlights the benefits of automation in reducing operational costs and improving decision accuracy.

Inference:
The paper emphasizes automation and supplier evaluation but lacks integration with real-time monitoring and penalty mechanisms. The proposed system fills this gap by combining evaluation, monitoring, and AI-based recommendations into a unified platform.

Paper 5: Application of Artificial Intelligence in the Supply Chain: A Systematic Literature Review (2022/2023)

Abstract:
This paper explores the role of AI in supply chain management, including demand forecasting, logistics optimization, and risk management. It highlights how AI can improve coordination, reduce delays, and enhance overall system efficiency.

Inference:

While the paper focuses on supply chain optimization, it indirectly supports the importance of AI in contract execution and monitoring. However, it does not address tender-specific challenges. The proposed system adapts these concepts to procurement by introducing post-award monitoring and delay-based penalty recommendations.

2.2 Books / Articles / Newspaper Referred

Various books, online articles, and industry reports were referred to understand procurement systems and AI applications:

- Government procurement guidelines and e-procurement frameworks
- Articles on digital transformation in procurement systems
- Documentation of AI tools such as Scikit-learn and NLP frameworks
- Case studies from organizations implementing e-tendering systems

These sources helped in understanding real-world challenges, system requirements, and implementation strategies for intelligent procurement platforms.

2.3 Interaction with Domain Experts

Insights were gathered by understanding the workflow followed by procurement officers and committee members involved in tender evaluation and contract monitoring.

Key observations include:

- Bid evaluation involves multiple parameters and requires careful comparison
- Manual processes lead to delays and inconsistencies
- Difficulty in identifying fraudulent bidders
- Lack of structured monitoring after contract allocation

2.4 Patent Search

Patent 1: Contract Lifecycle Management (AI-based)

Title: Contract Lifecycle Management

Year: 2022

Summary: This patent describes an AI-driven system that extracts contract data, analyzes KPIs, and automates contract creation, renewal, and modification. It also uses OCR + machine learning + blockchain for security.

Link: (https://patents.google.com/patent/US11348352B2)

Patent 2: System for Management of Lifecycle of Contracts

Title: System and Method for Management of Life Cycle of Contracts

Year: 2023

Summary: This patent explains the full lifecycle: RFP → negotiation → drafting → approval → execution. It highlights the need for automation due to large contract volumes.

Link: (https://patents.justia.com/patent/20230229866)

Patent 3: Electronic Contract Management System

Title:Electronic Contract Management Method and Device

Year: 2020

Summary: This patent focuses on digital contract storage, processing, and automation using electronic systems.

Link: (https://patents.google.com/patent/CN111724155A/en)

Patent 4: Bidding and Contract Management System

Title: Internet-Based Bidding and Contract Management System

Year: 2003

Summary: This patent introduces an online system for *tender bidding + contract handling*, espcially in construction projects. Link: (https://patents.google.com/patent/CN1402856A/en

Chapter 3: Requirement of Proposed System

This chapter presents the detailed requirements of the proposed **Intelligent Tender Management and Contract Lifecycle Platform**. The system is designed using an incremental development approach, allowing it to be built in modules such as tender management, AI-based evaluation, fraud detection, and chatbot assistance. These requirements are derived from the analysis of existing systems, literature survey, and the implementation planning discussed in the project.

3.1 Functional Requirements

The functional requirements describe the core functionalities that the system must perform.

The system shall provide secure user registration and login functionality for both Admin and Bidders. It shall implement role-based access control to ensure that different users have appropriate permissions. The Admin shall be able to create and publish tenders with all necessary details, while the system shall also allow updating, deleting, and viewing of tenders.

Bidders shall be able to submit bids along with required documents, and the system shall validate these submissions by preventing duplicate, incomplete, or late entries. The system shall include an AI-based evaluation module that uses machine learning models to evaluate bids based on weighted parameters and generate rankings automatically.

To enhance transparency, the system shall include fraud detection using techniques such as Isolation Forest and Random Forest to identify suspicious bidding patterns. It shall also assign risk scores to bidders based on their behavior and historical data.

The system shall support contract monitoring by tracking project progress and identifying delays after the tender is awarded. Additionally, it shall include a RAG-based chatbot using LLaMA to assist users with queries related to penalties, delays, and tender processes.

The system shall integrate all modules through REST APIs connecting the frontend, backend, AI services, and database. It shall also generate reports and analytics to support decision-making and performance tracking.

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and performance standards of the system.

The system shall provide fast processing and near real-time responses to ensure efficient user experience. It shall ensure strong security through authentication mechanisms and data protection techniques. The system shall be reliable, operating with minimal downtime and errors.

It shall be scalable to handle a large number of users and increasing volumes of data. The system shall be user-friendly, with an intuitive interface designed using modern UI/UX principles. It shall be available 24/7 as a web-based application.

Maintainability is also important; therefore, the system shall be designed using a modular architecture that allows easy updates and extensions. Additionally, the system shall ensure data integrity by maintaining accurate and consistent information throughout all operations.

3.3 Constraints

The proposed system is subject to several constraints.

It must comply with government procurement rules and policies, which may limit certain functionalities. The availability of real-world tender datasets is limited, which may affect the training and accuracy of AI models.

The project is also constrained by time and budget limitations as it is being developed within an academic framework. The system depends on a stable internet connection for proper functioning.

Furthermore, AI models used in the system may not always produce perfectly accurate results, which is a limitation of current machine learning techniques.

3.4 Hardware and Software Requirements

The system requires standard hardware and software configurations for development and deployment.

On the hardware side, a system with an Intel i5 processor or higher, a minimum of 8 GB RAM, and at least 256 GB storage is required. A stable internet connection is necessary for accessing and running the system.

On the software side, the frontend is developed using React.js along with HTML, CSS, and JavaScript. The backend is built using Node.js and Express.js. The database used is MongoDB or PostgreSQL. Programming languages used include JavaScript and Python. Development tools such as Visual Studio Code and GitHub are used for coding and version control.

3.5 Techniques Utilized

The proposed system uses several modern techniques for efficient functioning.

Machine learning is used for bid evaluation and scoring. Anomaly detection techniques such as Isolation Forest and Random Forest are used for fraud detection. Natural Language Processing (NLP) is used to develop the chatbot interface.

The system also uses Retrieval-Augmented Generation (RAG) with LLaMA to provide intelligent and context-aware chatbot responses. Data analytics techniques are used for tracking performance and generating insights. The entire system is developed as a web-based application.

3.6 Tools Utilized

Various tools are used in the development of the proposed system.

Frontend development is done using React.js along with HTML, CSS, and JavaScript. Backend development is carried out using Node.js and Express.js. MongoDB or PostgreSQL is used as the database.

Development tools include Visual Studio Code and GitHub for coding and version control. AI and machine learning functionalities are implemented using Python along with libraries such as Scikit-learn and TensorFlow.

3.7 SDGs Mapped with the Proposed Project and Justification

The proposed system aligns with

Sustainable Development Goal 16: Peace, Justice and Strong Institutions.

Traditional tendering systems often lack transparency and are prone to corruption and inefficiencies. The proposed platform addresses these issues by introducing AI-based automation and digital workflows.

The system ensures transparency through digital record-keeping and automated processes. It promotes fair decision-making by using AI-based bid evaluation and ranking. Fraud and corruption are reduced through anomaly detection techniques.

Additionally, the system improves institutional efficiency by automating procurement processes and reducing manual intervention. Therefore, the proposed system contributes to building strong, transparent, and accountable institutions.

CHAPTER 4: PROPOSED DESIGN

This chapter describes the design of the proposed Intelligent Tender Management and Contract Lifecycle Platform. It explains the system architecture, modular structure, design diagrams, algorithms used in existing systems, and project scheduling. The design follows an incremental and modular approach as presented in the project PPT.

4.1 Block Diagram Representation of the Proposed System with Explanation

The block diagram of the proposed system represents the overall architecture and interaction between different components. The system consists of four main layers: Frontend, Backend, AI Services, and Database.

The frontend is developed using React.js and provides the user interface for Admin and Bidders. It allows users to perform actions such as registration, login, tender creation, and bid submission.

The backend, built using Node.js and Express.js, handles business logic and API requests. It acts as an intermediary between the frontend and other components. All system operations such as validation, authentication, and request processing are handled here.

The AI services layer is implemented as a Python microservice. It includes machine learning models such as Isolation Forest and Random Forest for fraud detection and bid evaluation. It also integrates a RAG-based chatbot using LLaMA for intelligent query handling related to penalties and delays.

The database layer uses MongoDB to store user data, tender details, bids, and contract information. All modules interact with the database through the backend.

Overall, the block diagram shows a complete flow where the frontend communicates with the backend, which in turn interacts with AI services and the database to process and return results.

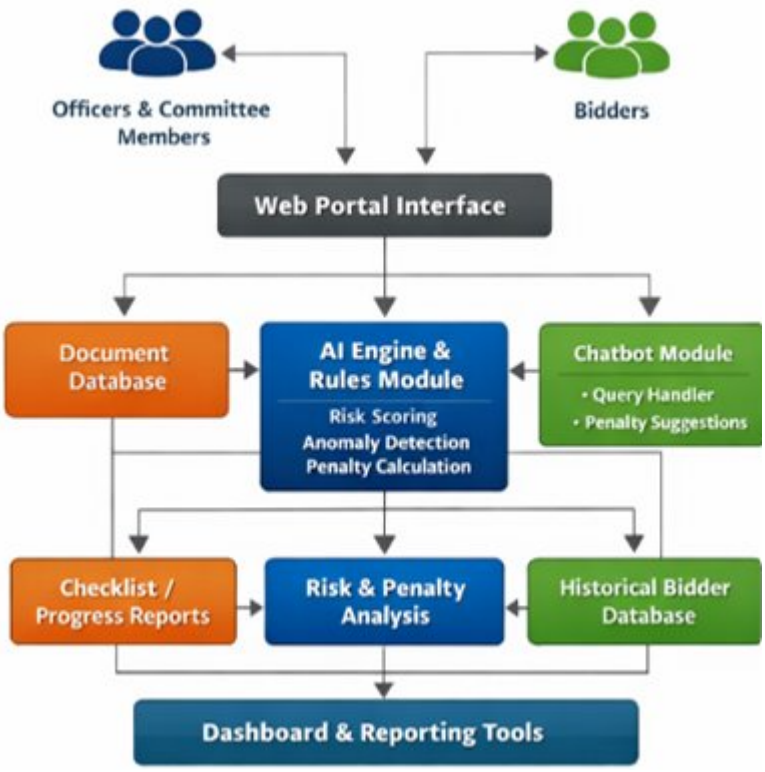


Fig 1: Block diagram

4.2 Modular Diagram Representation of the Proposed System with Explanation

The modular diagram divides the system into smaller functional units or modules for better organization and development.

The User Management Module handles registration, login, and role-based access control. The Tender Management Module is responsible for creating, updating, and publishing tenders.

The Bid Management Module allows bidders to submit bids and ensures validation of submissions. The AI Evaluation Module processes bids using machine learning models to evaluate and rank them.

The Fraud Detection Module identifies suspicious bidding patterns using anomaly detection techniques. The Contract Management Module tracks the progress of awarded tenders and monitors delays.

The Chatbot Module provides assistance to users using NLP and RAG-based techniques. The Reporting Module generates analytics and reports for decision-making.

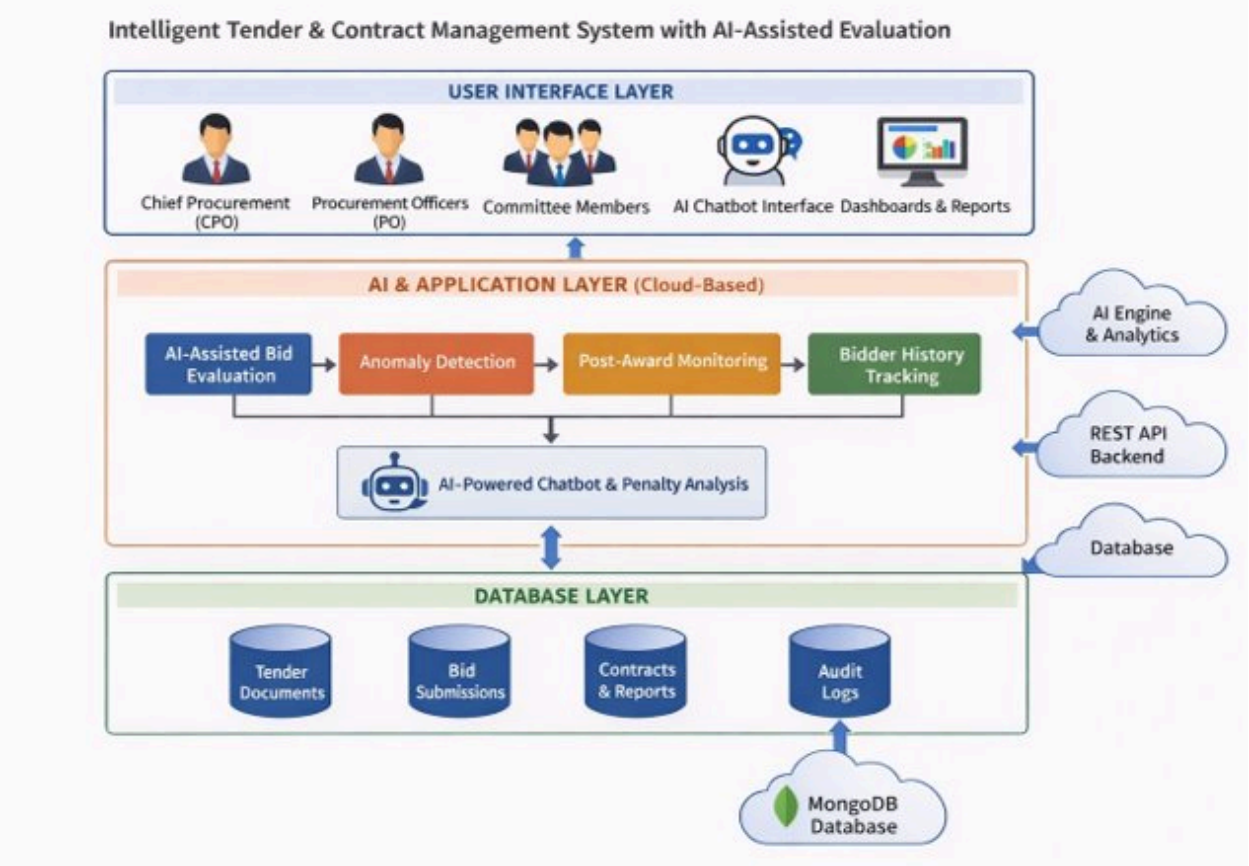


Fig 2: Modular Diagram

4.3 Design of the Proposed System

Data Flow / Use Case Diagram

The data flow in the system begins when a user logs into the platform. Admin users create tenders, which are stored in the database and made available to bidders. Bidders view tenders and submit bids along with necessary documents.

These bids are sent to the backend and further processed by the AI evaluation module. The system calculates scores, detects anomalies, and generates rankings. The results are then stored and displayed to the Admin.

Post-award, contract monitoring begins, where progress reports are submitted and tracked. The chatbot assists users by answering queries related to tenders, penalties, and delays.

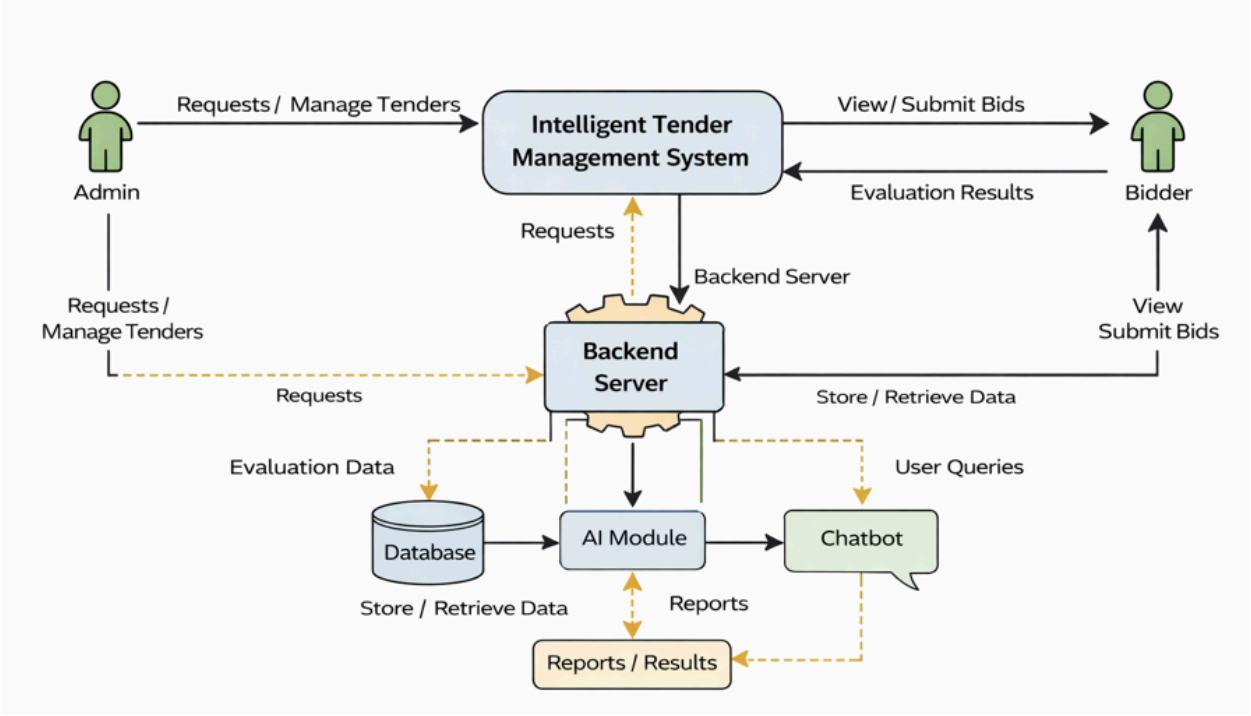


Fig 3: Data Flow Diagram (Level 0)

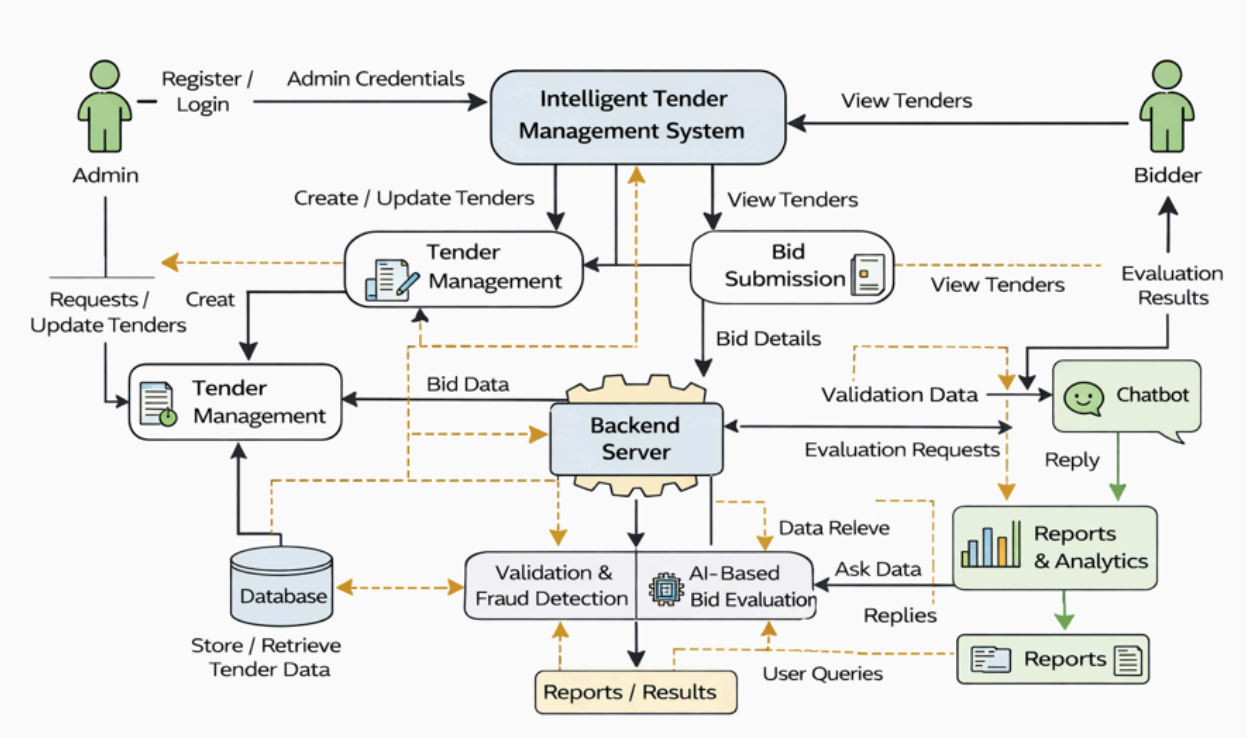


Fig 4: Data Flow Diagram (Level 1)

Flowchart

The system flow starts with user authentication. If the user is an Admin, they can create or manage tenders. If the user is a Bidder, they can view tenders and submit bids.

Once a bid is submitted, it undergoes validation. Valid bids are sent to the AI evaluation module, where scoring, ranking, and fraud detection are performed. The system then displays results and allows further contract monitoring.

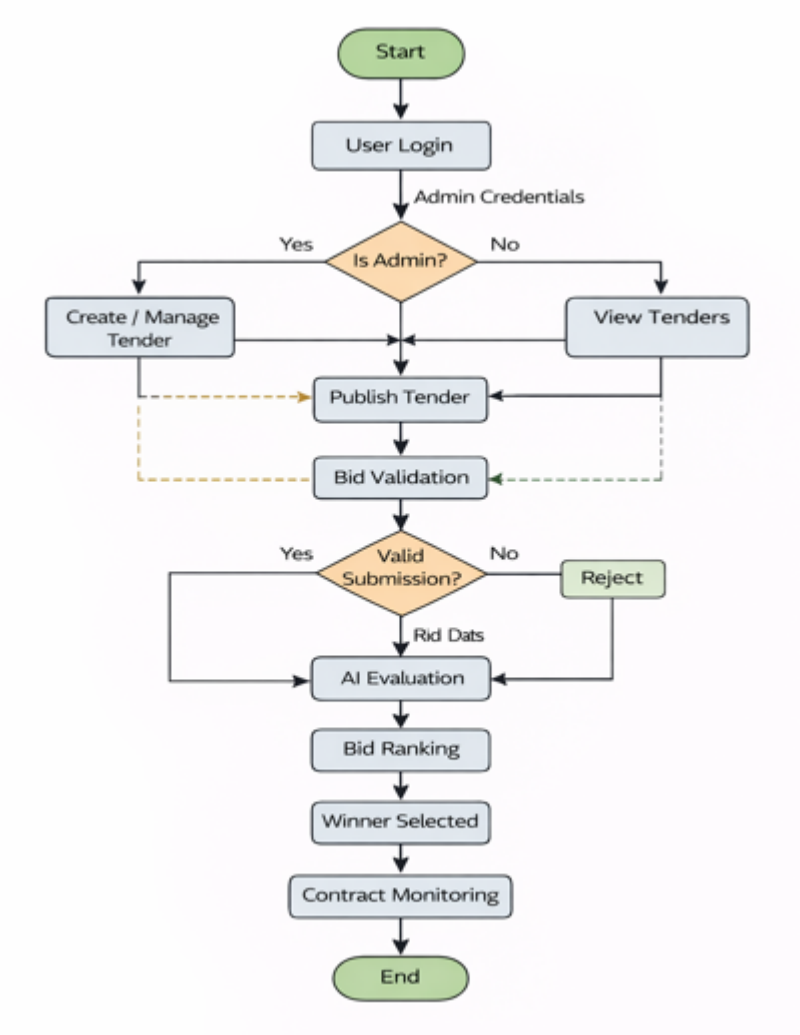


Fig 5: Flowchart

Activity Diagram / State Transition

The system transitions through multiple states such as user login, tender creation, bid submission, evaluation, and contract monitoring.

For example, a tender moves from "Created" to "Published," then to "Bidding Open," followed by "Evaluation," and finally to "Awarded." After awarding, it enters the "Monitoring" state where progress is tracked.

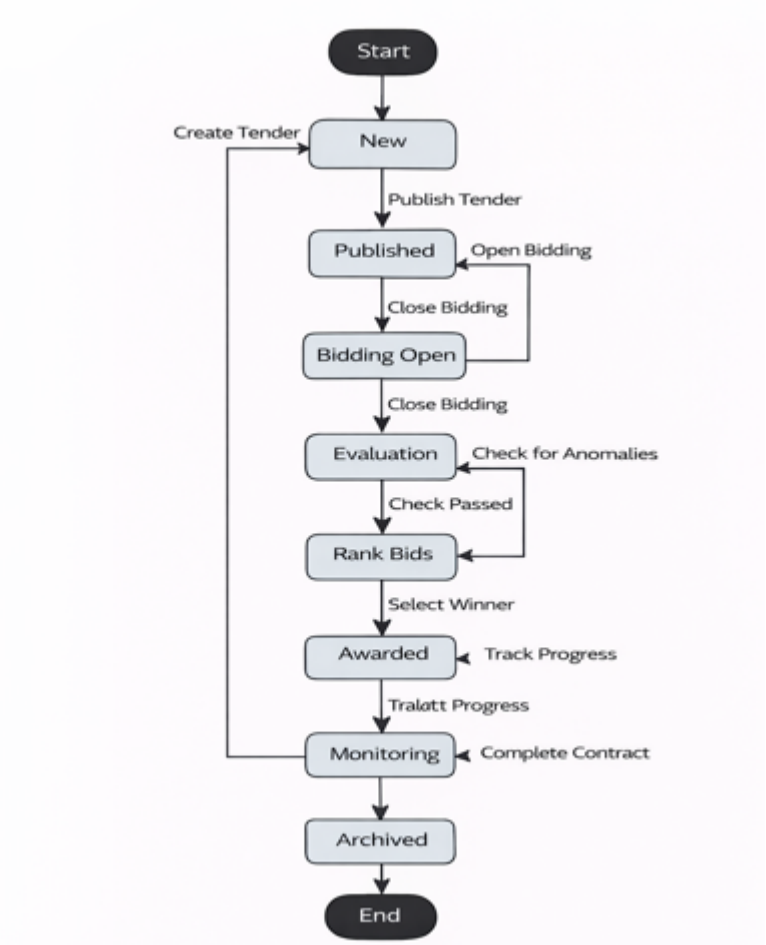


Fig 6 : State Diagram

ER Diagram

The Entity-Relationship (ER) design includes key entities such as User, Tender, Bid, Contract, and Report.

A User can be an Admin or Bidder. Admins create tenders, while bidders submit bids. Each bid is linked to a specific tender. Once a bid is accepted, a contract is created. Reports are generated based on tender and contract data.

This design ensures proper relationships and efficient data management in the system.

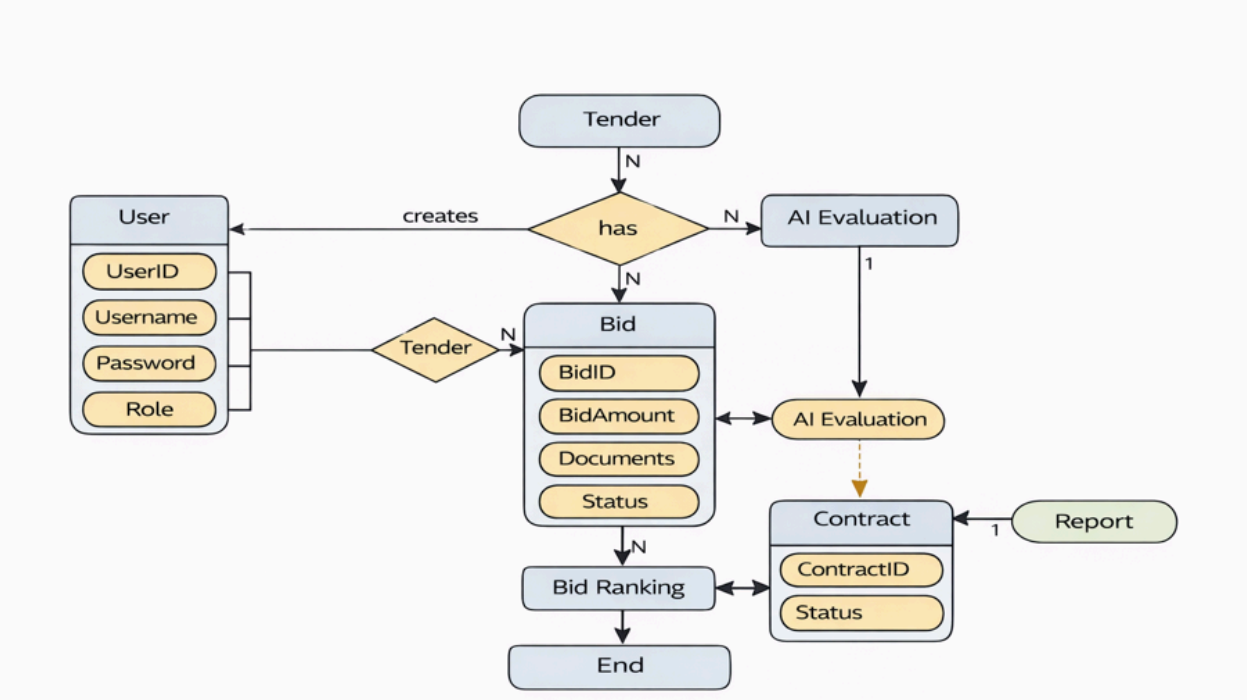


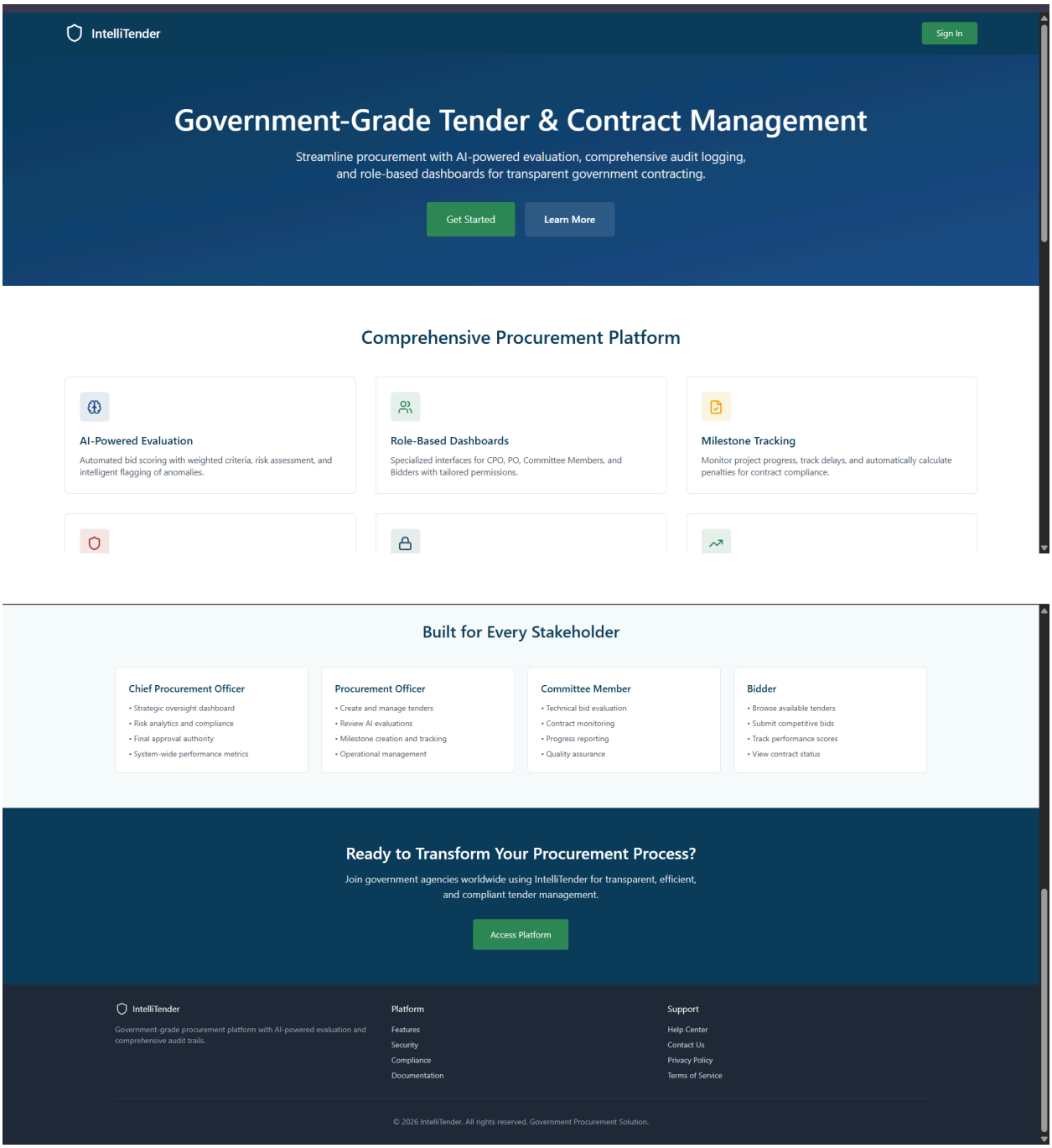
Fig 7: ER Diagram

Screenshot of Implementation

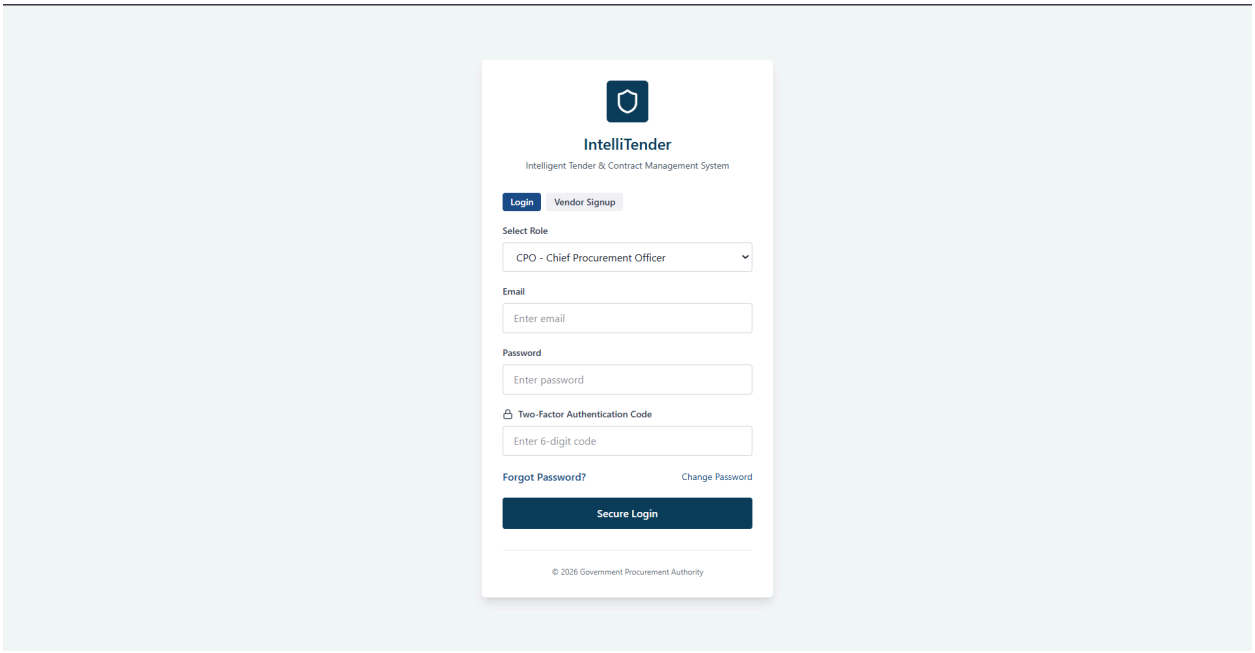
The implementation includes UI designs created using Figma and partial development of modules such as MongoDB schema design and frontend interfaces.

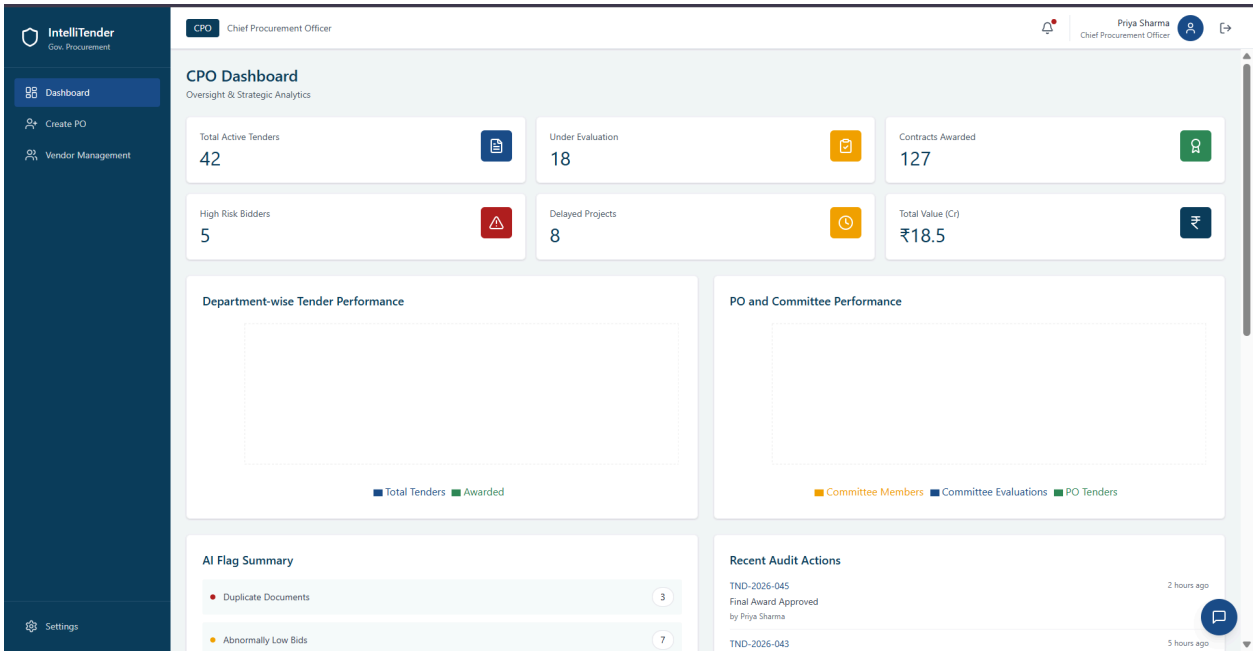
Screenshots include:

- User interface design (Figma)
- Database schema (MongoDB)
- Initial implementation of system modules

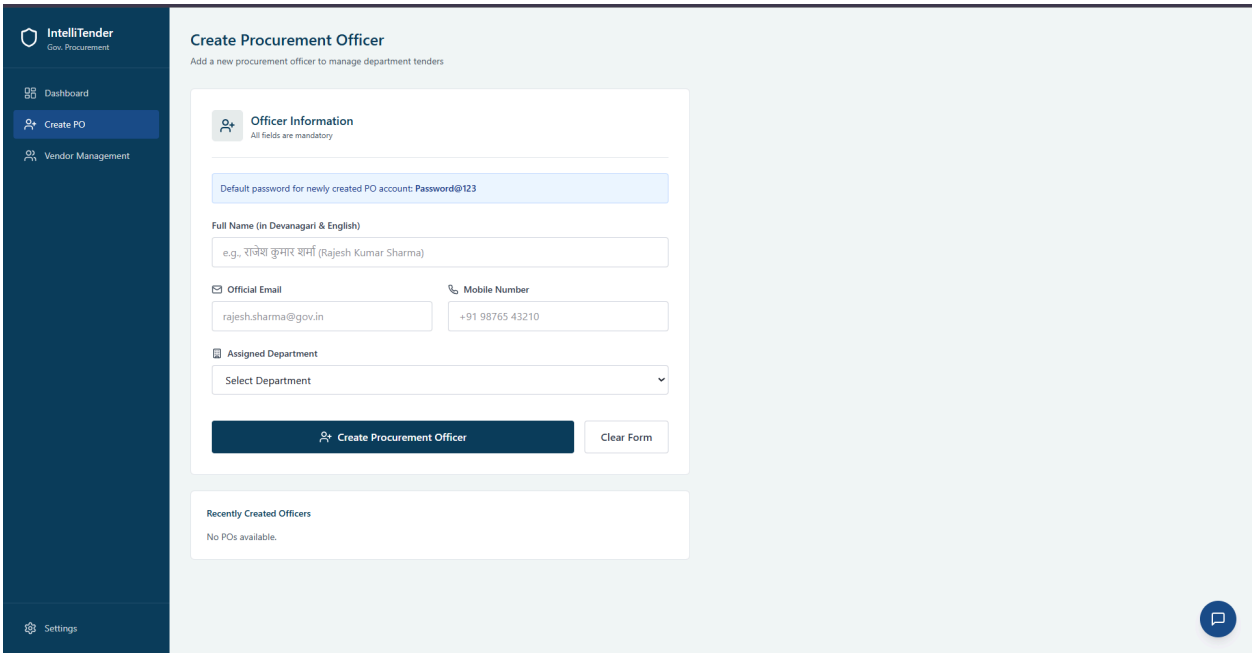


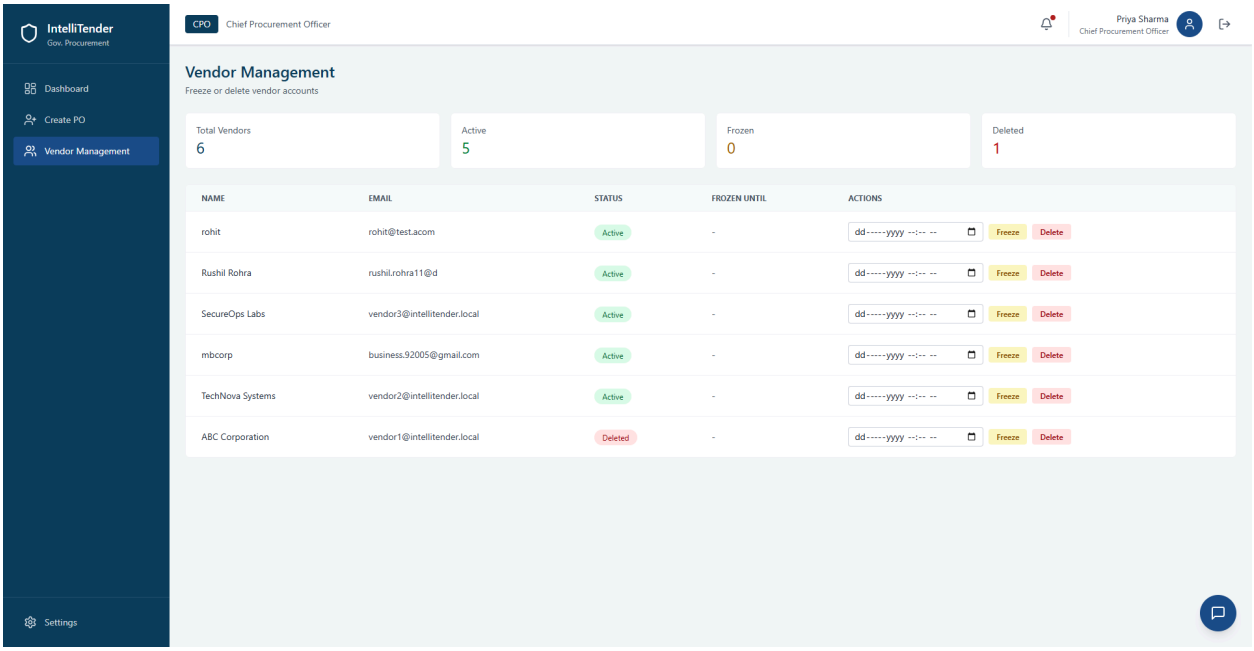
Home page / dashboard



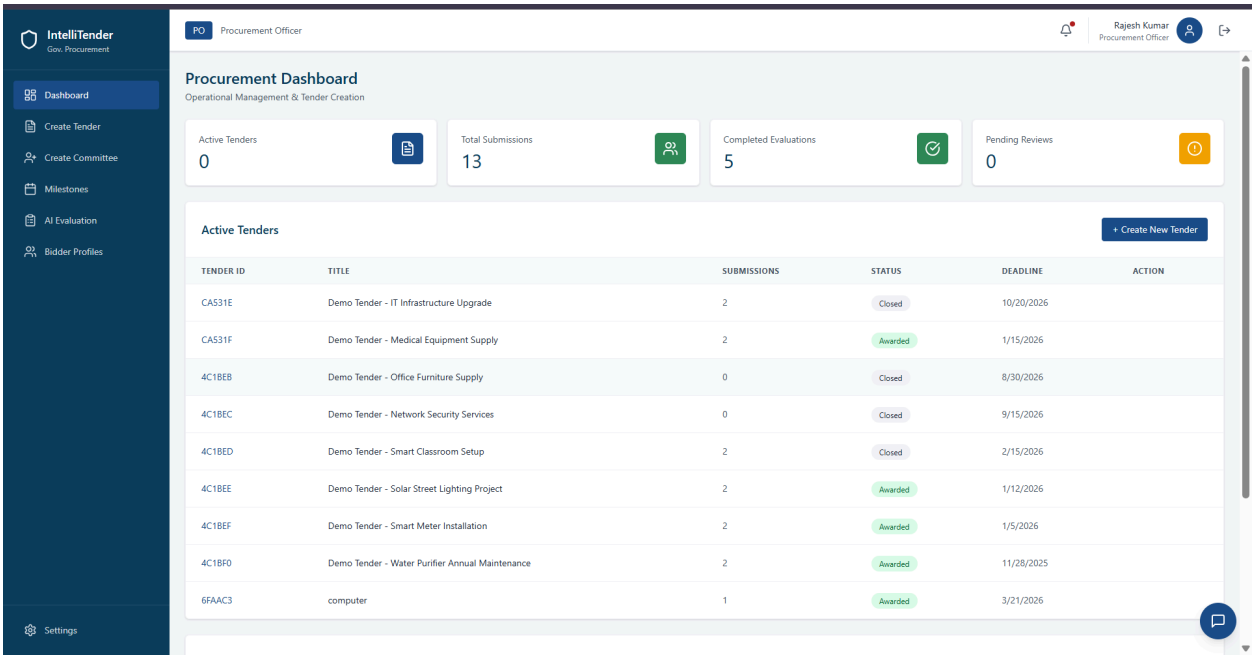


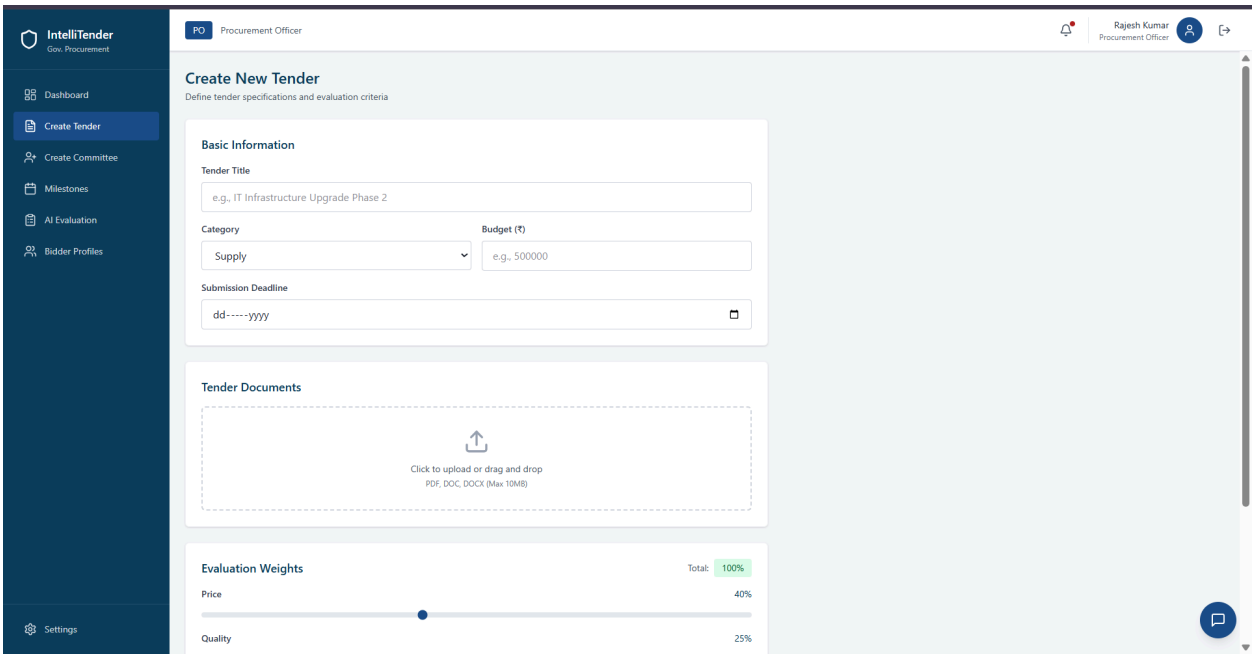
Login page and Dashboard page



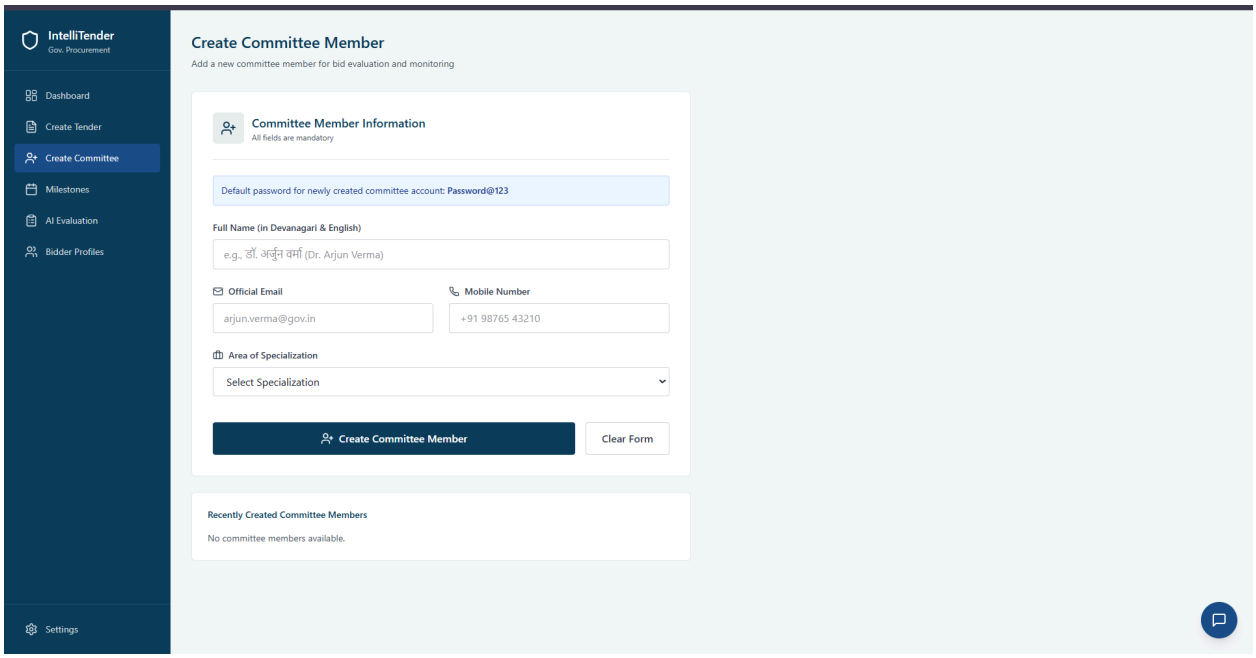


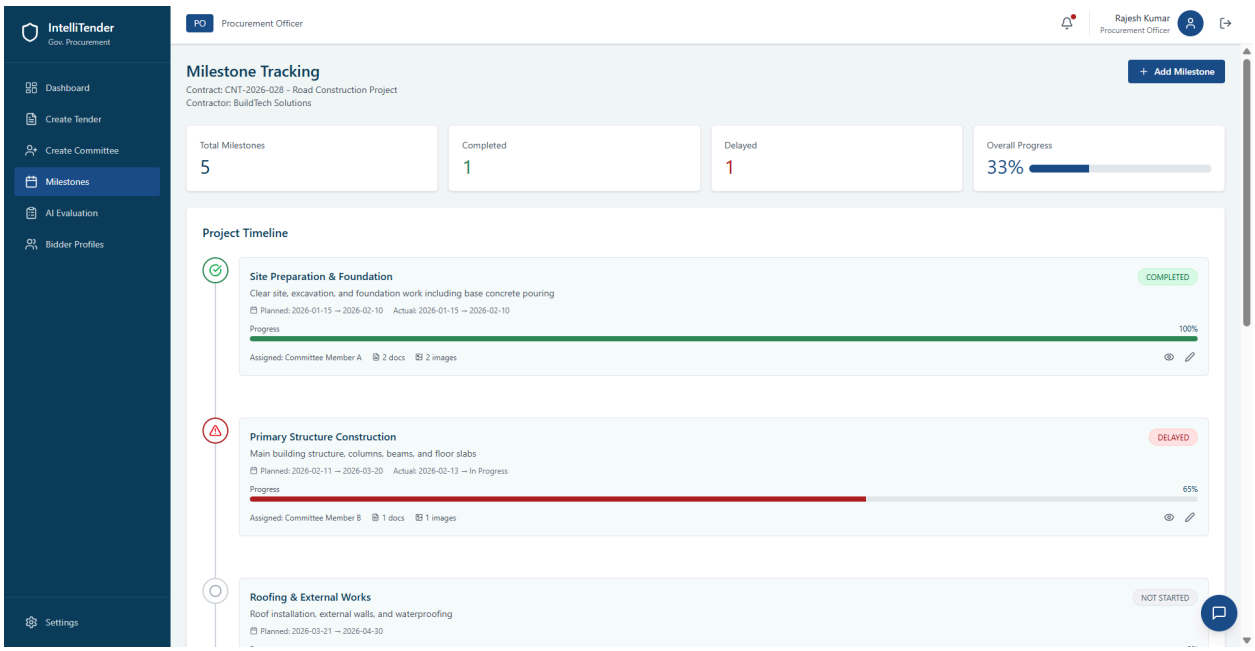
CPO Dashboard





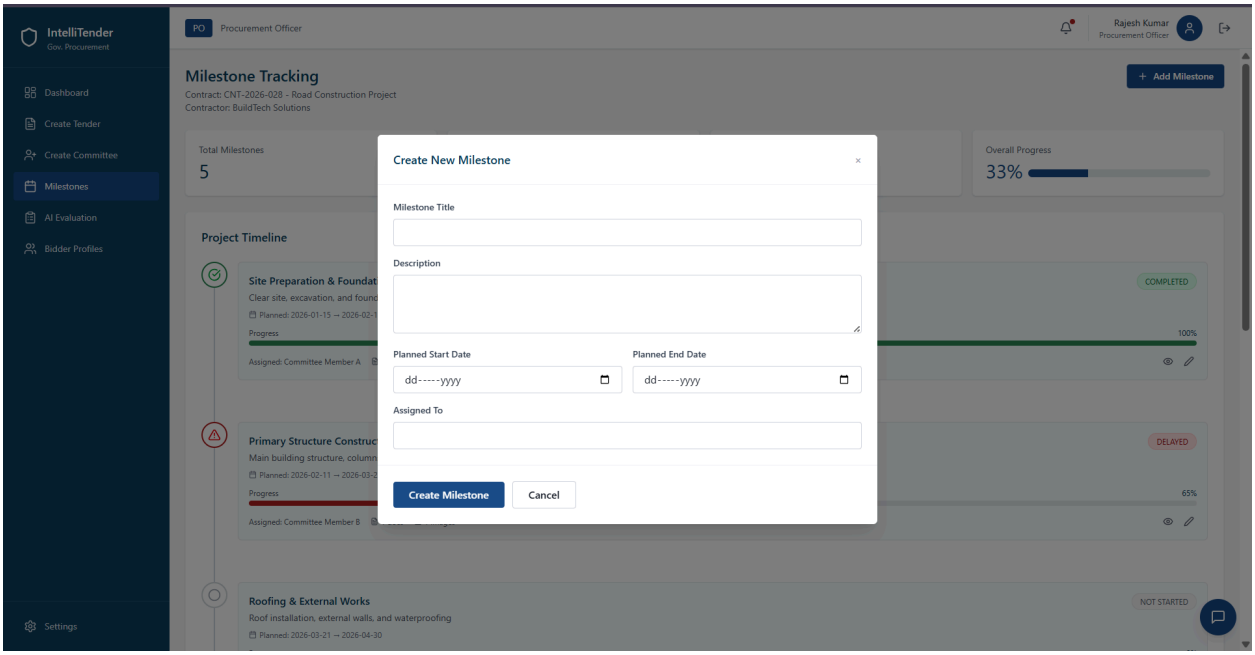
PO Dashboard

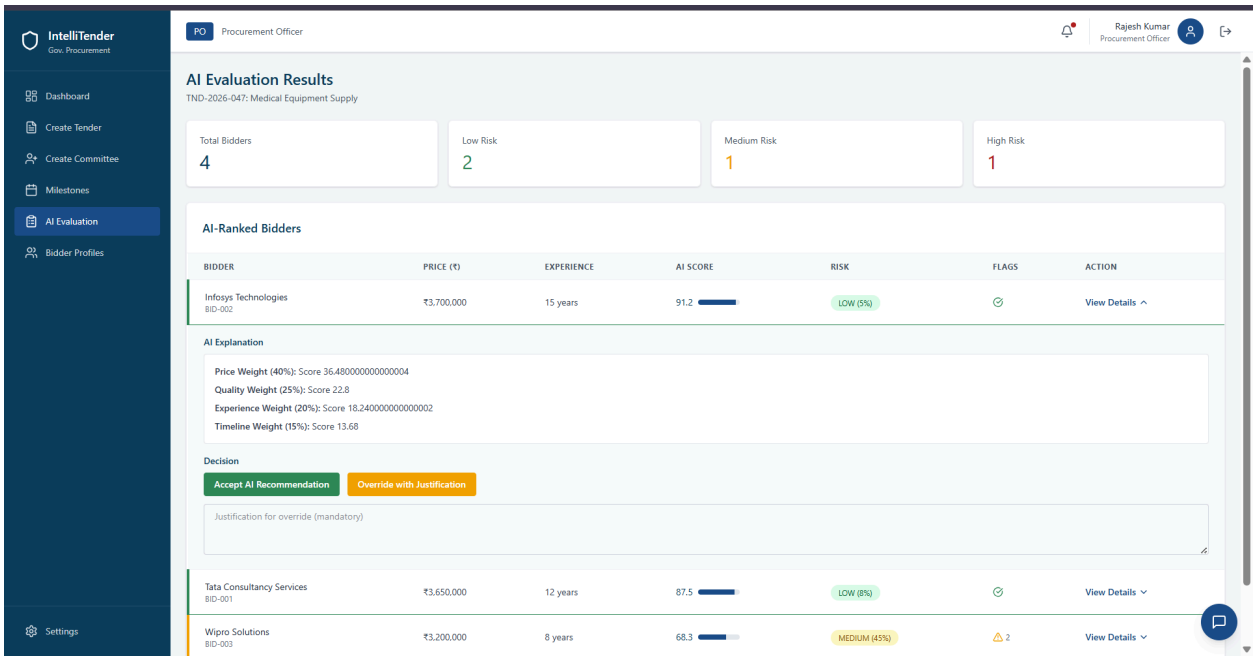


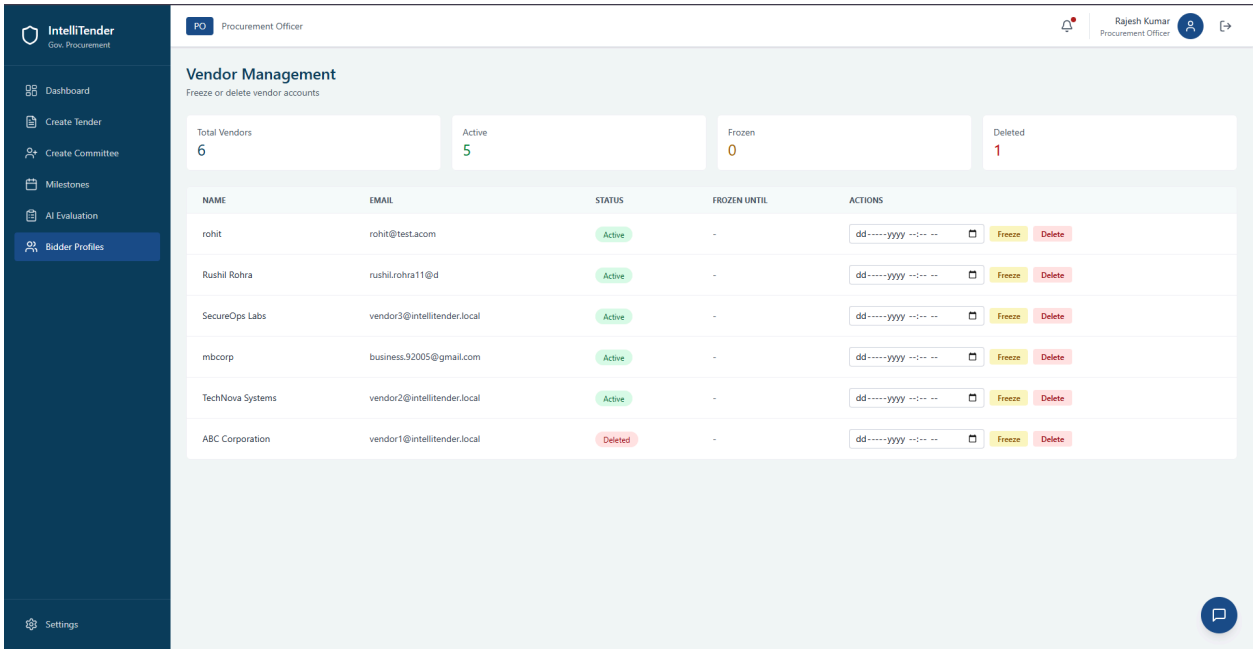


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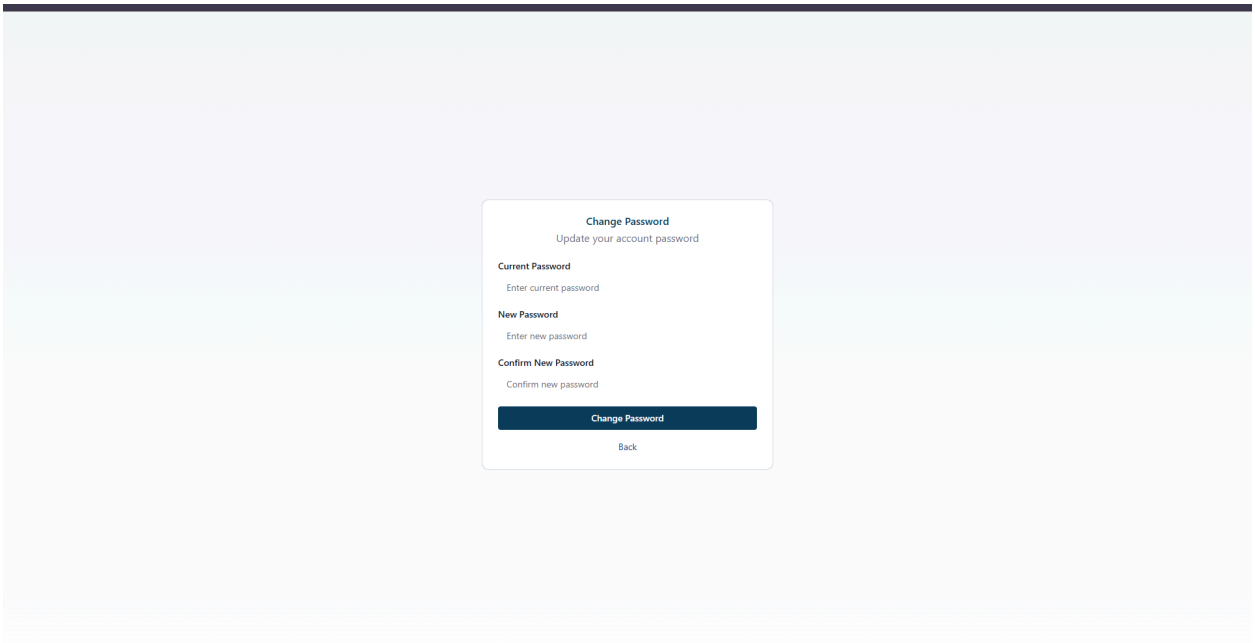
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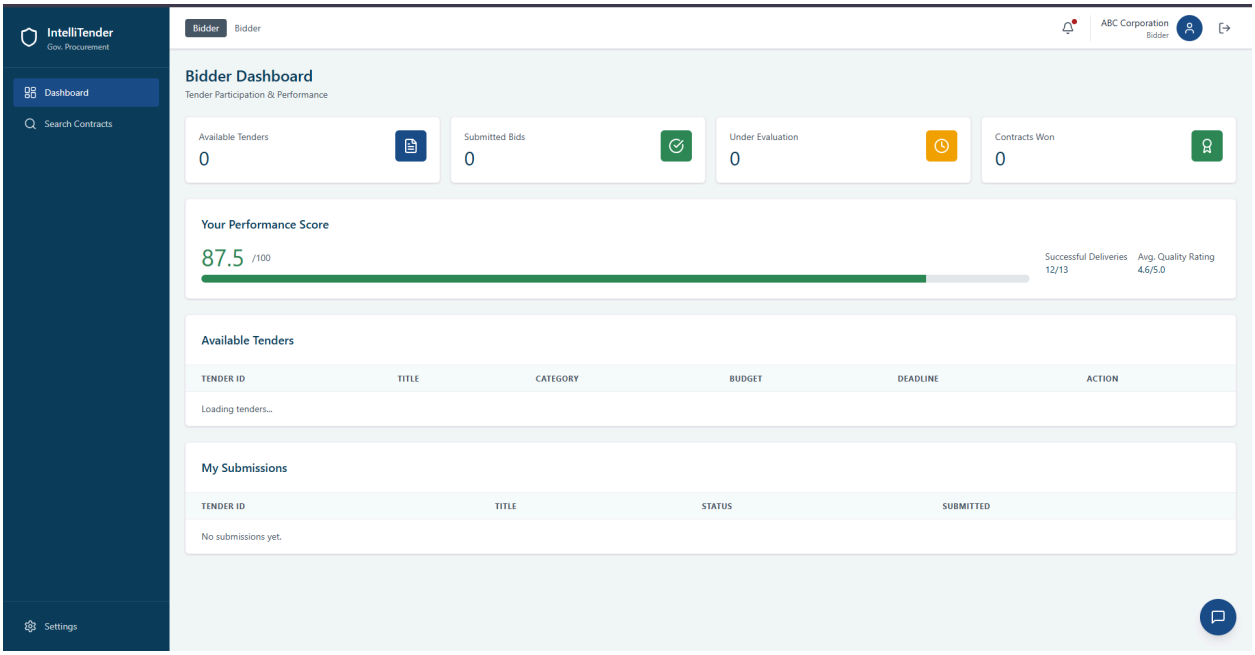




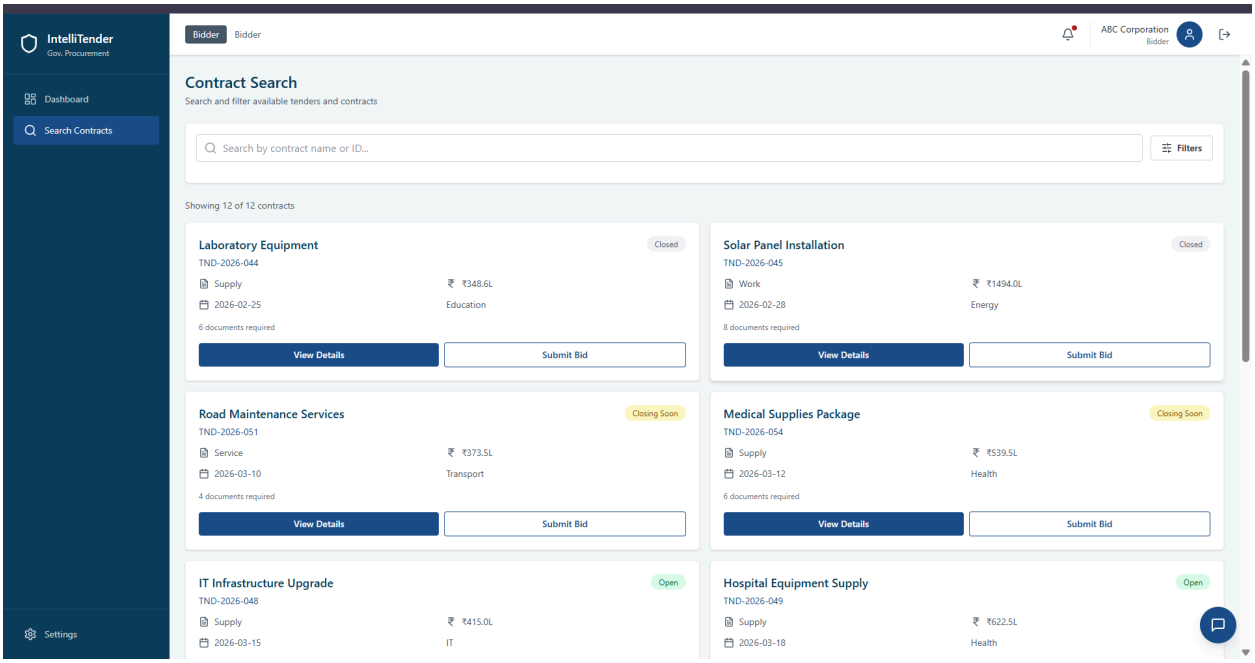
PO Dashboard



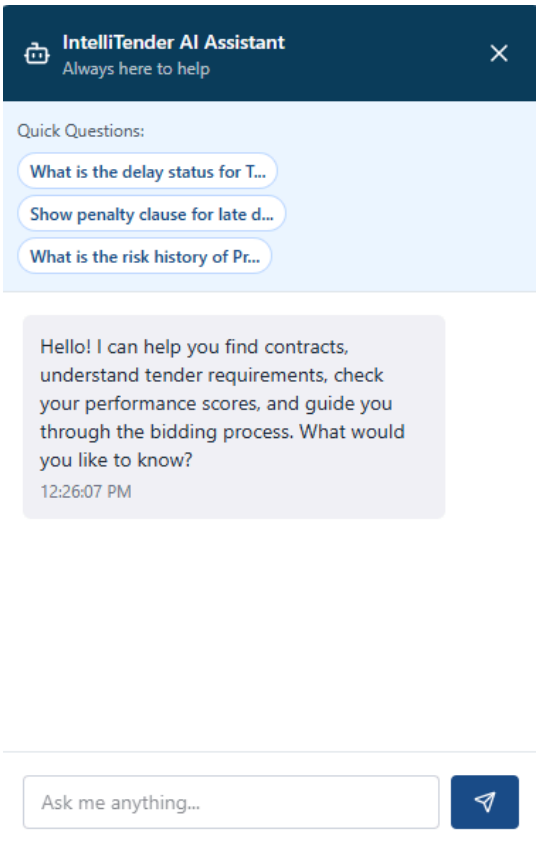
Forgot password page



Vendor Dashboard



Vendor Dashboard



AI Chatbot

4.4 Algorithms Utilized in Existing Systems

Existing procurement systems use basic algorithms for bid evaluation, such as lowest-cost selection or rule-based filtering. These methods are simple but lack intelligence and flexibility.

Some systems use statistical methods and basic machine learning models for supplier evaluation and fraud detection. However, they often do not support real-time analysis or advanced anomaly detection.

Compared to these, the proposed system improves performance by using advanced algorithms such as Random Forest and Isolation Forest for better accuracy and fraud detection.

4.5 Project Scheduling and Tracking using Timeline / Gantt Chart

The project follows an incremental development model, where tasks are divided into phases for systematic development and tracking.

The initial phase involved literature survey and problem identification. The next phase focused on UI/UX design using Figma. After that, database schema design and backend implementation were completed, including API development and integration with MongoDB.

Currently, the project is in the phase of AI model finalization and integration, including the use of Isolation Forest and Random Forest for anomaly detection and bid evaluation, as well as development of a RAG-based chatbot using LLaMA.

Future work includes integration of AI modules with the backend system, frontend-backend integration, system testing, performance optimization, and final deployment. Additional work will also focus on improving model accuracy and enhancing user experience.

Each phase is planned sequentially, similar to a Gantt chart, ensuring proper tracking of progress and timely completion of the project.

CHAPTER 5: IMPLEMENTATION, RESULTS AND DISCUSSION

5.1 Data Collection / Acquisition

The data for the proposed system is collected from multiple sources related to procurement processes. Since real-world tender data is confidential, synthetic and structured data is created to simulate actual scenarios.

The dataset includes bidder details, bid submissions, financial quotations, technical documents, and submission timestamps. Additional data is obtained from tender documents, which define evaluation criteria, weightages, contract clauses, milestones, and penalty rules.

For post-award monitoring, data is collected through checklist-based inputs (for supply contracts) and progress reports (for work contracts). The system also maintains bidder historical records such as past performance, delays, and penalties.

This data is used for training and testing AI models for bid evaluation, anomaly detection, and risk scoring.

5.2 Data Preprocessing Techniques

Before applying AI models, the collected data is preprocessed to ensure accuracy and consistency.

The preprocessing steps include data cleaning, removal of duplicate entries, and handling missing values. Numerical data such as bid amounts and timelines are normalized for better model performance.

Categorical data such as bidder type and contract type are encoded into machine-readable formats. Document-based inputs are processed using text extraction techniques for further analysis.

These preprocessing techniques improve the efficiency and accuracy of machine learning models.

5.3 Experimental Setup

The system is implemented as a web-based platform with integrated backend and database.

The backend is developed using Node.js and Express.js, and it is connected to MongoDB for data storage. API endpoints are created to manage user authentication, tender operations, and bid submissions.

AI models are implemented using Python libraries such as Scikit-learn. Techniques like Random Forest and Isolation Forest are used for bid evaluation and anomaly detection.

The system is tested using sample datasets to evaluate performance and functionality. The chatbot module is integrated using NLP techniques and RAG-based architecture for intelligent responses.

5.4 Evaluation Metrics

The performance of the proposed system is evaluated using the following metrics:

- **AI Risk Score Validity:** Measures the accuracy of detecting suspicious bids
- **Bid Evaluation Time Reduction:** Compares time taken with manual evaluation
- **Post-Award Monitoring Reliability:** Checks tracking of milestones and delays
- **Penalty Recommendation Correctness:** Verifies accuracy of AI-suggested penalties
- **User Adoption and Satisfaction:** Evaluates usability and system acceptance
- **Bidder Performance Profile Consistency:** Ensures correct historical tracking
- **Transparency and Audit Compliance:** Ensures proper logging and traceability

These metrics help in analyzing system efficiency, reliability, and effectiveness.

5.5 Gap Analysis of Existing Systems

Existing tender management systems mainly rely on manual processes and basic automation. They lack advanced features such as AI-based evaluation, fraud detection, and real-time monitoring.

Most systems do not detect anomalies like duplicate bids, shell companies, or abnormal pricing patterns. They also fail to integrate historical bidder performance into decision-making.

Post-award monitoring is often manual and inconsistent, leading to delays and lack of accountability. Additionally, there is no intelligent chatbot support for assisting users.

The proposed system addresses these gaps by integrating AI models, anomaly detection, chatbot assistance, and automated monitoring, resulting in a more efficient and transparent system.

CHAPTER 6: PLAN OF ACTION FOR THE NEXT SEMESTER

6.1 Work Done Till Date

Significant progress has been made in the development of the project.

The literature survey and problem analysis have been completed. The system design, including block diagram, modular diagram, and database schema, has been finalized.

The frontend UI/UX design has been created using Figma. The backend has been successfully implemented using Node.js and Express.js, along with API development and MongoDB integration.

Initial work on AI models such as bid evaluation and anomaly detection has been started. The overall system architecture and workflow have been clearly defined.

6.2 Plan of Action for Project II

In the next phase of the project, focus will be on completing and integrating all modules.

AI models for bid evaluation, fraud detection, and risk scoring will be finalized and integrated with the backend. The RAG-based chatbot will be fully implemented and connected to system data.

Frontend and backend integration will be completed to ensure smooth user interaction. System testing will be performed to check functionality, performance, and reliability.

Further improvements will include optimization of AI models, enhancement of user interface, and implementation of deployment using cloud platforms.

CHAPTER 7: CONCLUSION

The proposed **Intelligent Tender Management and Contract Lifecycle Platform** provides a modern solution to the limitations of traditional procurement systems.

By integrating AI-based bid evaluation, anomaly detection, and chatbot-assisted decision support, the system improves transparency, efficiency, and accountability. It reduces manual effort and helps in detecting fraudulent activities.

The inclusion of post-award monitoring and bidder history tracking ensures better contract management and informed decision-making.

Overall, the system provides a scalable and intelligent approach to procurement, making it suitable for real-world applications in government and corporate sectors.

CHAPTER 8: REFERENCES

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Appendix

a. List of Figures

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